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SkinFormer: a hybrid vision transformer and ConvNeXtV2 approach for skin cancer detection and segmentation

Yinyin Fu[1] & Chengjun Guo[2]✉

Skin cancer remains one of the most prevalent and deadly forms of cancer globally, where early detection plays a critical role in improving survival rates. In this study, we introduce SkinFormer, a novel hybrid deep learning architecture primarily designed for binary skin lesion classification (benign vs. malignant), enhanced by a standalone pretrained segmentation module for accurate ROI cropping. It incorporates Vision Transformers (ViT), ConvNeXtV2, and a Separable Self-Attention method to improve the precision and interpretability of skin lesion classification and segmentation. Utilizing a portion of the ISIC 2024 dataset alongside the DermQuest dataset, which includes more than 55,000 images for binary classification of benign and malignant skin lesions, SkinFormer attains promising performance with an accuracy of 96.8%, an AUC of 0.985, and an F1-score of 95.8%. The independent segmentation module attains a Dice Similarity Coefficient of 95.6% and an Intersection over Union of 91.8%. The approach effectively resolves issues including irregular lesion borders, low contrast, and artifacts through the integration of global contextual reasoning and local texture extraction. Ablation experiments validate the importance of each architectural module, while qualitative heatmaps and overlay visualizations enhance its clinical interpretability. Statistical significance was established by McNemar's test ($p < 0.01$), underscoring the method's resilience. The results indicate that SkinFormer is a potential instrument for automated skin cancer diagnosis, a potential tool for clinical decision support and teledermatology applications.

Keywords Cutaneous Carcinoma, Deep Learning, Vision Transformer, ConvNeXtV2, Lesion Segmentation, Medical Image Analysis, Explainable Artificial Intelligence

Skin cancer remains a major global health challenge, motivating the development of large-scale dermoscopicdatasets such as ISIC and HAM10000, which enable robust benchmarking and advancement of automatedmelanoma detection methods based on computational analysis[1,2,3,4]. Melanoma, a particularly severe form of skin cancer, arises from uncontrolled proliferation of melanocytes and is strongly associated with ultraviolet (UV) radiation exposure, hereditary factors, and environmental influences, such as prolonged sunlight or tanning bed use. Skin cancer, arising in the epidermis, is amenable to early visual detection, and prompt diagnosis can markedly improve treatment results by preventing swift spread to critical organs such as the brain, liver, and lungs[5–7]. Recent biomedicalstudies highlight the role of molecular pathways such as PI3K/AKT/mTOR and ferroptosis in cancer progression andtreatment, emphasizing the importance of advanced analytical and intelligent diagnostic approaches for early detection andintervention[15]. Furthermore, recent clinical studies indicate that genetic mutations and viral factors significantly increasecancer risk and progression, reinforcing the need for reliable and data-driven diagnostic systems for early detection and riskassessment[16]. The diagnostic process is complicated by variability in lesion appearance, artifacts like hair, and the subjective nature of dermatological evaluations, which vary with experience, yielding accuracies of approximately 80% for dermatologists with over 10 years of practice compared to 62% for those with 3–5 years. Dermoscopy has historically been essential for the non-invasive examination of skin lesions, enhancing visibility beyond the capabilities of the unaided eye[10]. Nonetheless, its efficacy depends on clinical expertise[11], highlighting the need for automated, objective diagnostic tools[12,13].

[1]Department of Ophthalmology, Sichuan Provincial People's Hospital, University of Electronic Science and Technology of China, Chengdu 610072, China. [2]School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China. ✉email: johnsonguo@uestc.edu.cn

In standard clinical practice, the diagnosis of skin cancer is often reliant on a mix of visual examination, dermoscopy, and histological evaluation. Clinical examination employs established criteria like asymmetry, border irregularity, color variation, and lesion diameter, whereas dermoscopy facilitates the observation of subsurface structures and pigmentation patterns imperceptible to the human eye. Notwithstanding these advancements, histological examination of biopsy specimens continues to be the definitive standard for diagnosis. In recent years, image-based computational analysis has developed into a significant decision-support method, designed tooffer objective and reproducible evaluations that enhance clinical knowledge. Recent studies have demonstrated theeffectiveness of deep learning in enhancing medical imaging quality and resolution, particularly in complex diagnosticscenarios such as ultrasound microvessel imaging, where super-resolution techniques significantly improve the visibility offine structures and diagnostic accuracy[14]. Recent advances in intelligent control and reinforcement learning have shown strong capabilities in modeling complex,nonlinear, and dynamic systems, enabling adaptive decision-making under uncertainty. In particular, approaches based onadaptive dynamic programming and event-triggered control have demonstrated effectiveness in optimizing performance andensuring system stability in time-delay and multi-agent environments[8,9].

Early machine learning techniques, including support vector machines (SVM), logistic regression, and random forests, havebeen employed to classify skin lesions; however, their reliance on manual feature engineering and limited ability to handlehigh-dimensional dermoscopic data hinder their scalability and generalizability. Moreover, recent studies have revealed thatdeep learning-based image analysis systems are vulnerable to adversarial attacks and data perturbations, while advancedcomputational models such as low-rank learning and kernel-based methods have been explored to enhance robustness anduncover complex associations in biomedical data, improving the reliability of intelligent diagnostic systems[17,18]. The emergence of deep learning, particularly convolutional neural networks (CNNs) like GoogleNet, ResNet, DenseNet, and EfficientNet, has improved skin cancer detection; however, these models necessitate large datasets and considerable computational resources, often leading to prolonged processing times that are unfeasible in resource-constrained settings. Acoustic interference from hair or variable pigmentation hinders accurate segmentation, as architectures like U-Net struggle to define multi-scale lesion boundaries in restricted datasets[19]. The InSiNet model[20], employing a CNN and Inception-based U-Net, mitigated certain restrictions and attained an accuracy of 94.59% on the 2018 ISIC dataset. Nonetheless, its dependence on conventional preprocessing and segmentation methods indicates potential for enhancing accuracy and efficiency.

This study introduces SkinFormer, a novel hybrid architecture that integrates ViT and ConvNeXtV2 to enhance diagnostic precision and computational efficacy. The system features an advanced preprocessing pipeline that employs edge-aware filtering and deep adaptive thresholding, alongside Swin Transformer-based segmentation, achieving an IoU exceeding 0.92, thereby surpassing the limitations of traditional methods like Gaussian blur and U-Net. This study seeks to overcome these limitations by employing diverse, modern datasets such as ISIC 2024 and DermQuest to develop a robust model. This research is important as it may provide dermatologists with an automated, interpretable tool for early skin cancer detection, reducing reliance on expert experience, enhancing diagnostic precision, and enabling application in resource-limited environments. The principal accomplishments of this study are as follows:

- SkinFormer is introduced as an innovative hybrid model that adeptly integrates the Vision Transformer backbone and ConvNeXtV2 with a Separable Self-Attention fusion mechanism, facilitating precise classification and segmentation of skin lesions.
- A thorough preprocessing pipeline is introduced, incorporating edge-aware filtering and deep adaptive thresholding, which markedly improves lesion localization and model generalization across artifacts and varying image conditions.
- Through rigorous quantitative and qualitative evaluations, SkinFormer is shown to outperform state-of-the-art models on the ISIC 2024 and DermQuest datasets, achieving statistically significant improvements in performance and clinical interpretability.

Unlike existing approaches that solely utilize transformer architectures or convolutional networks, SkinFormer explicitly integrates hierarchical local feature extraction with global contextual reasoning inside a unified dual-branch framework. The proposed system incorporates a separable self-attention fusion mechanism that enables efficient inter-branch interaction while maintaining computational scalability, unlike previous hybrid models. The close integration of Swin Transformer-based segmentation with the classification backbone addresses a significant limitation of prior methods that regard segmentation and classification as loosely connected tasks, ensuring that diagnostically pertinent lesion areas are consistently emphasized.

The following portions of this document are structured as outlined below: Sect. 2 outlines the materials and methods, detailing the ISIC 2024 and DermQuest datasets, preprocessing techniques, segmentation approach, and SkinFormer architecture. Section 3 outlines the experimental setup, including training, validation, and testing methods, along with performance metrics like as accuracy, sensitivity, specificity, and computing time. Section 4 elucidates and analyzes the results, contrasting SkinFormer with premier models in the domain. Section 5 analyzes the implications, limitations, and potential pathways. Section 6 concludes with a synthesis of contributions and clinical ramifications.

Related works

Deep learning techniques have significantly improved the classification and segmentation of skin lesions. In particular, hybrid models that mix Transformer topologies with Convolutional Neural Networks (CNNs) have shown better performance in these tasks. For example, an explainable hybrid deep learning framework for accurate skin lesion segmentation and multi-class classification was presented in a recent paper[21]. This model improves lesion boundary identification accuracy, a critical problem in clinical dermatology, by combining CNNs and BiLSTM (Bidirectional Long Short-Term Memory). Even when complicated variables like uneven boundaries and poor contrast are present, the hybrid technique improves the model's capacity to reliably categorize and segment lesions.

Another study[22] investigated a hybrid strategy integrating ResNet50V2 with Swin Transformer for magnetic resonance imaging-based brain tumor classification in the field of preprocessing for lesion classification. The methods employed for effective feature extraction and segmentation show promise for enhancing skin lesion analysis, particularly in clinical imaging contexts, even if this study is not directly connected to skin lesions.

Another noteworthy method for classifying skin lesions was reported in[23], which used CNN and BiLSTM in an attention-based hybrid approach. By utilizing attention mechanisms, which enable the model to concentrate on important regions of the image, this model successfully enhances skin lesion categorization, improving interpretability and accuracy.

In the field of dermoscopy and clinical imaging melanoma detection, the Vision Transformer (ViT) model has been increasingly utilized due to its capacity to capture global context within images. A study[24] introduced DermViT, a ViT-based model for robust and efficient skin lesion classification. The model demonstrated superior performance over traditional CNN-based models by incorporating global context, which improved classification accuracy, especially for melanoma detection.

Additionally, combining CNNs with Transformer models has proven beneficial for lesion boundary detection. A model introduced in[25] used CNN-Transformer fusion networks for lesion segmentation, with a particular focus on improving boundary detection. The fusion of boundary sensing and Transformer-based contextual modeling helped refine the segmentation process, which is critical for accurately delineating skin lesions in dermoscopic images.

In another work[26], a sequential approach was adopted to combine segmentation and classification tasks. The model first segments the lesion and then classifies it, ensuring that the preprocessing step closely aligns with the classification task. This sequential learning approach highlights the importance of preprocessing in enhancing the overall performance of skin lesion classification systems, particularly when dealing with complex lesion structures.

Another study[27] introduced ESC-UNET, a hybrid CNN and Swin Transformer model specifically designed for skin lesion segmentation. This model demonstrated the ability to accurately segment lesions, including those with complex boundaries, by leveraging both CNNs and Transformers in a unified architecture. The integration of Swin Transformer allowed for better handling of various lesion types and sizes, leading to improved segmentation results.

The use of multi-level attention for skin lesion segmentation was explored in[28], where the SkinAttn-Net model was introduced. This model utilizes multi-level attention mechanisms to focus on both local and global image features, allowing it to more effectively segment skin lesions. By using attention at multiple levels, the model can prioritize critical regions in the image, thus improving the segmentation of lesions with diverse textures and structures.

In the area of interpretability and model transparency, another framework[29] employed SHAP (SHapley Additive exPlanations) to explain the decision-making process of deep learning models for skin cancer classification. The use of SHAP values allows for a clearer understanding of how the model makes predictions, which is crucial in medical applications where interpretability is essential for clinical trust.

Another study[30] proposed a state space modeling approach combined with convolutional perception for skin lesion segmentation. This model integrated mathematical modeling with deep convolutional networks to enhance segmentation accuracy. By incorporating state space models, the approach improved the efficiency and accuracy of skin lesion segmentation, demonstrating the potential of combining traditional mathematical techniques with deep learning for medical image analysis.

A signaling entropy-based approach for evaluating customized route activation in survival analysis using PanCancer data is proposed in the paper[31]. It supports deep learning imaging models in risk assessment and diagnostic customization by concentrating on scoring pathway activation in various individuals, which aids in predicting clinical outcomes in a variety of malignancies, including skin cancer. Although ConvNeXt-ST-AFF[32] successfully integrates ConvNeXt and Swin Transformer via an Attention Feature Fusion (AFF) module for adaptive weighted aggregation of branch features, it lacks explicit decoupling of spatial and channel dependencies and depends on implicit blending via attention-guided weights. Additionally, it conducts generic multi-class skin disease classification without ROI cropping to concentrate on lesion-specific regions or segmentation-guided preprocessing. On the other hand, SkinFormer presents a structured separable self-attention fusion that enables explicit and effective inter-branch interaction by first sequentially modeling spatial dependencies and then refining them channel-wise.

The independent pretrained Swin Transformer segmentation module offers effective ROI cropping without the need for joint training, mitigating artifact sensitivity and border inconsistency more efficiently than conventional preprocessing or loosely integrated hybrids. These design decisions directly address significant remaining constraints restricted long-range modeling in CNNs, loss of local detail in transformers, and suboptimal fusion in current hybrids while ensuring computational scalability[33].

In[34], a novel lightweight CNNs architecture named EMDA-Net (Earth Motion Distance (EMD)-Influenced Attention-Assisted Neural Network) was proposed, specifically tailored for cancer subtype classification. Their model architecture is based on MobileNetV2 and incorporates integrated attention mechanisms supported by statistical methods. By using the strengths of MobileNetV2 and attention mechanisms, it achieves great performance while preserving a lightweight design. To enhance similarity detection in data channels, they introduce an innovative attention strategy influenced by EMD. The growing utilization of artificial intelligence (AI) in dermatology necessitates the assurance of equity in the creation of machine learning models, particularly in skin lesion classification, where decisions can profoundly affect individuals' lives. The study[35] examines biases among various Fitzpatrick skin types in baseline pre-trained models and assesses diverse training methodologies to mitigate these discrepancies. An unsupervised skin transformation is created to modify the Fitzpatrick skin type (FST) of an image, utilizing joint regularization and synthetic image fusion techniques to mitigate bias problems. Furthermore, explainable artificial intelligence (XAI) methodologies, such gradient-weighted class activation mapping (Grad-CAM), are employed to discern the fundamental sources of bias within the models.

Materials and methods

The SkinFormer framework provides a holistic, end-to-end solution for skin cancer analysis by consolidating data preparation, lesion identification, and diagnostic inference under a unified transformer-centric paradigm. Commencing with a meticulously selected collection of dermoscopic images sourced from the ISIC 2024 and DermQuest repositories, each instance undergoes automated removal of hair artifacts by edge-aware filtering and deep adaptive thresholding. The refined image is next examined by a Swin Transformer module that generates high-fidelity, multi-scale masks outlining the lesion. The masks guide the cropping of the region of interest, which is reduced to 224×224 pixels and normalized to the [0,1] intensity range before being fed into the SkinFormer backbone. In SkinFormer, a Vision Transformer branch gathers long-range contextual information, while a ConvNeXtV2 branch proficiently extracts local texture and color characteristics; removable self-attention layers enable feature fusion across several scales. A definitive global pooling operation compresses the fused representation into a fixed-length vector, while an efficient fully connected layer with sigmoid activation produces probabilistic scores for benign and malignant classification. The entire network is trained end-to-end using the Adam optimizer with binary cross-entropy loss, and its diagnostic performance is evaluated on reserved validation and test sets through accuracy, sensitivity, specificity, and ROC AUC, followed by statistical analysis to confirm the model's robustness and clinical relevance. Figure 1 depicts an overview of the planned architecture.

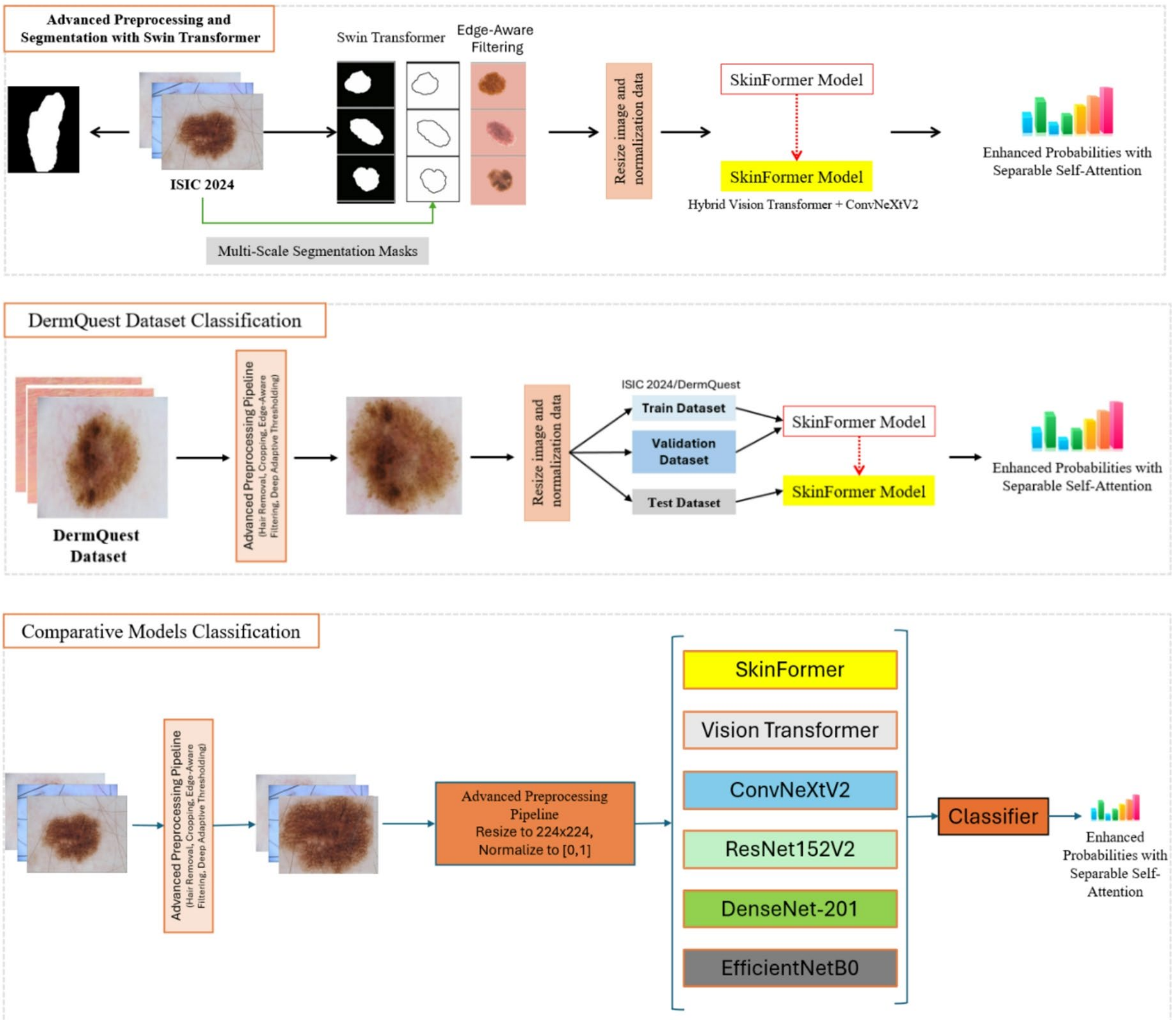

Fig. 1. overview of the proposed architecture.

Dataset	Task	Total Images (used)	Class Balance Description
ISIC 2024 (subset)	Binary (Benign/Malignant)	~ 40,000	Highly imbalanced (malignant rare)
DermQuest	Binary (Benign/Malignant)	~ 15,000	Moderately balanced
Combined	Binary (Benign/Malignant)	> 55,000	Clinically realistic imbalance

Table 1. Summary of dataset composition and class balance.

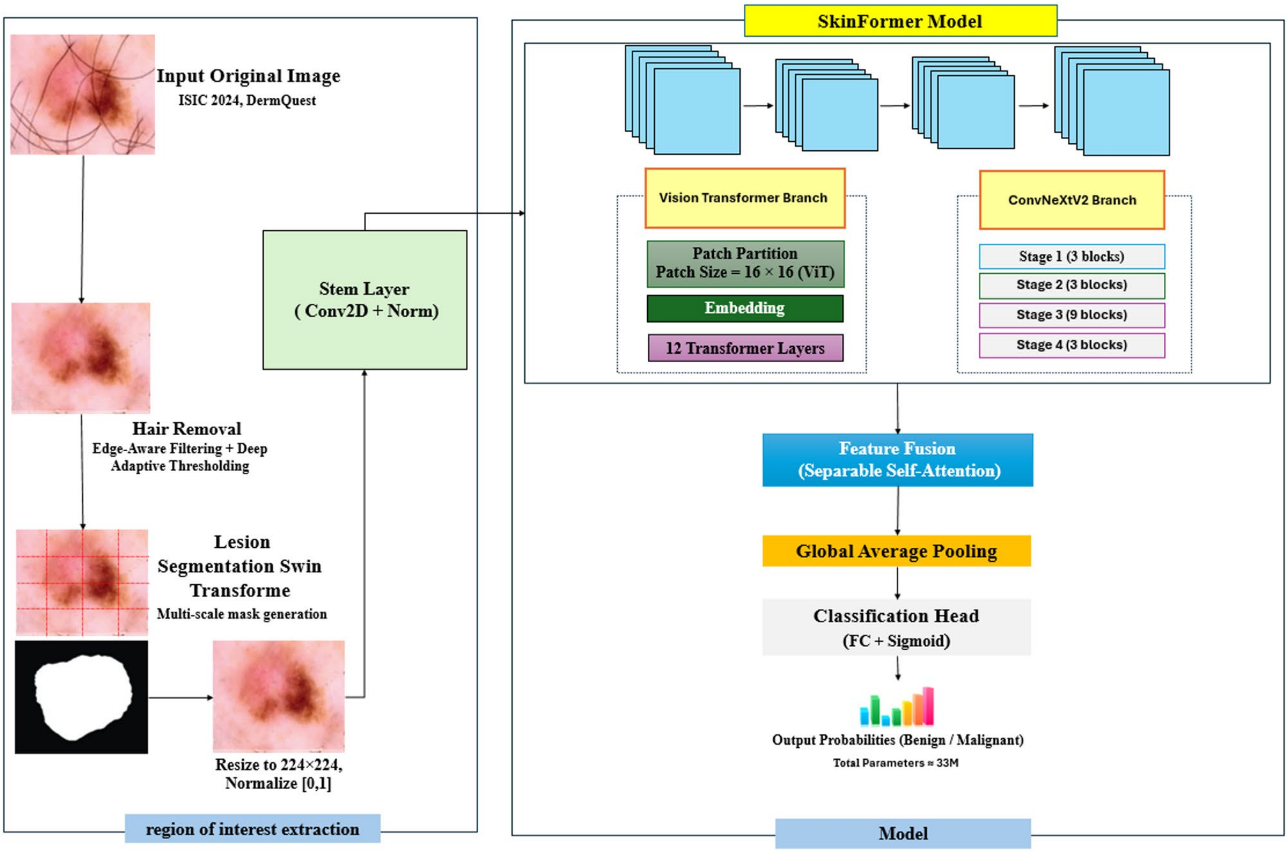

Fig. 2. The framework of the proposed approach.

Dataset

A comprehensive picture library for skin lesion analysis was created by integrating a portion of the ISIC 2024 challenge dataset with the DermQuest dataset. The complete ISIC 2024 dataset comprises over 401,000 cropped lesion images derived from 3D total body photos, emphasizing binary classification (benign versus malignant). To guarantee computational viability and equitable training while preserving clinical authenticity, we chose a representative selection of over 40,000 photos from ISIC 2024. The DermQuest collection adds around 15,000 pictures, largely classified into benign and malignant categories.

Table 1 summarizes that both datasets are intended for binary classification, exhibiting considerable class imbalance due to the rarity of malignant cases, which mirrors real-world prevalence. The aggregated corpus surpasses 55,000 photos, offering significant heterogeneity in lesion morphology, imaging modalities, and anatomical sites.

All experiments reported in this study are conducted exclusively on this combined dataset of over 55,000 images. No additional external test sets were used.

No images were eliminated because of incomplete metadata, corrupted files, missing labels, missing annotations, or other problems with the quality of the data. The selected subsets from the ISIC 2024 challenge dataset (about 40,000 images) and the DermQuest dataset (roughly 15,000 images) were used in full following application of the described preprocessing strategy. There were no missing values in the class labels (benign/malignant) or any given metadata.

Preprocessing pipeline

Prior to the integration of dermoscopic images into the SkinFormer backbone, all inputs undergo a three-stage preprocessing pipeline, as seen in (Fig. 2) on the left. This pipeline comprises (1) hair/artifact removal with standard techniques, (2) lesion segmentation through an independent Swin Transformer module, and (3) ROI cropping and normalization, guaranteeing artifact removal, accurate lesion delineation, and consistent picture dimensions/intensity. The initial phase employs edge-aware filtering and deep adaptive thresholding to eliminate hair and artifacts (e.g., ruler marks, ink, bubbles) without any deep learning components.

The second stage employs a standalone pretrained Swin Transformer module independent of the classification backbone and not jointly trained to generate a high-fidelity binary lesion mask. This module is used solely for preprocessing and does not contribute to the final classification output. The third stage uses this mask to crop the region of interest (ROI), centered on the lesion with a 28-pixel border removal to avoid peripheral artifacts, resized to 224×224 pixels and normalized to [0,1] before input to the classification backbone.

Lesion segmentation

The lesion segmentation is executed by a discrete pretrained Swin Transformer module, functioning separately from the SkinFormer classification backbone. This module is not optimized in conjunction with end-to-end training and is solely utilized to produce a binary lesion mask for region of interest cropping in the preprocessing pipeline. The input is segmented into non-overlapping 4×4 patches, which function as tokens for self-attention. Hierarchical stages utilizing shifted-window multi-head self-attention (window size 7×7) effectively capture both local texture and global context, yielding a high-fidelity binary mask (foreground: lesion; background: healthy skin). The shifted window attention layers enable cross-window interactions that represent the lesion's global context in relation to the surrounding skin.

The network is pretrained using a comprehensive collection of dermoscopic segmentation masks, and its multi-scale feature hierarchy produces a binary mask in which foreground pixels signify the lesion and background pixels indicate healthy tissue. In minimally invasive surgery, a cooperative framework for surgical tool segmentation with an emphasis on monocular depth estimate is introduced[36]. An adversarial method is used to create a few-shot segmentation model of medical pictures based on bidirectional augmented Mamba[37]. This technique attains segmentation accuracy exceeding 0.92 IoU on designated validation data and adeptly handles lesions of varying size, shape, and contrast through the seamless integration of local and global cues. Consult Fig. 3 for the detailed illustration of the Swin Transformer block.

The proposed skinformer architecture

SkinFormer is a hybrid deep learning framework consisting of two main components: (1) a preprocessing and lesion segmentation pipeline that generates a cropped region of interest (ROI) through an independent Swin Transformer-based segmentation module, and (2) a dual-branch classification backbone (Vision Transformer branch + ConvNeXtV2 branch) employing separable self-attention fusion for binary classification (benign versus malignant). The model obtains a preprocessed dermoscopic image measuring $224 \times 224 \times 3$. The initial stem layer implements a 4×4 convolution with a stride of 4, succeeded by layer normalization, converting the input into a patchified feature representation with diminished spatial dimensions (56×56) while maintaining low-level visual structures. Subsequent to this stem layer, the network diverges into two complimentary branches: a Vision Transformer (ViT) encoder and a ConvNeXtV2 encoder. In the ViT branch, the feature map is divided into non-overlapping 16×16 patches, flattened, and linearly transformed into token embeddings. The tokens traverse 12 transformer layers (ViT-B configuration), each comprising multi-head self-attention (6 heads) and a feed-forward MLP. Windowed and shifted-window self-attention techniques, inspired by the Swin Transformer, are integrated to improve local-global interaction and computational efficiency. The Swin-inspired attention is incorporated within the ViT blocks and is completely separate from the independent Swin Transformer segmentation module utilized in preprocessing.

To enhance local-global interaction, we specify that the 16×16 patch size pertains solely to the ViT encoder. The Swin-based modules utilized in preprocessing and segmentation implement distinct hierarchical window sizes and patch-merging techniques, independent of the ViT patch size. The ConvNeXtV2 branch emulates the hierarchical convolutional design of modern CNNs, comprising four stages with depths {3, 3, 9, 3} and channel widths varying from 96 to 768. Every stage utilizes depthwise separable convolutions, layer normalization, GELU activation, and pointwise convolution, enabling multi-scale extraction of texture-sensitive features. The outputs from the two branches are concatenated and subsequently analyzed by a feature fusion module utilizing separable self-attention. This fusion approach enables inter-branch interaction while reducing computational

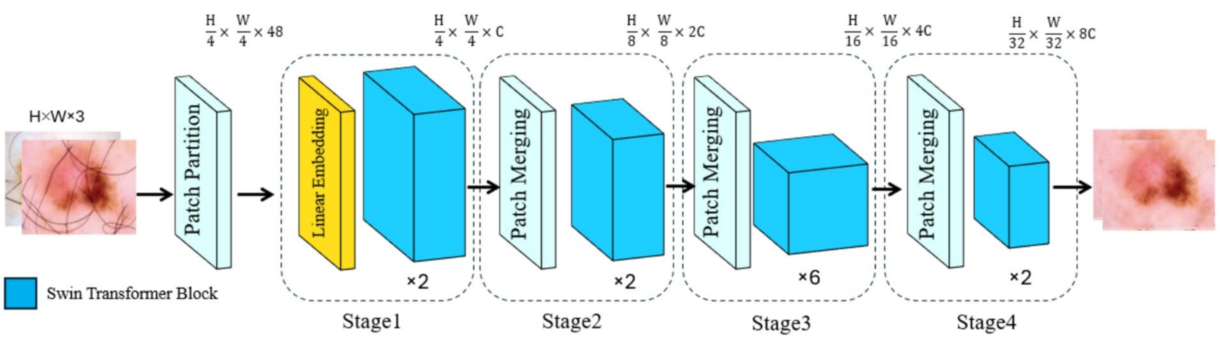

Fig. 3. Swin Transformer block.

Parameter	SkinFormer Value	Notes
Batch Size	32	
Epoch	50 (max, with early stopping)	Cosine annealing schedule
Optimizer	AdamW	
Loss Function	Binary Cross-Entropy	For classification task; segmentation module pretrained separately
Random State	42	
Regularizer	L2 + Dropout (0.2)	Dropout applied in classification head
Learning Rate	5e-05	
Transformer Layers	12	ViT-B configuration
Window Size (Swin)	7 × 7	For windowed attention in ViT branch
Attention Heads	6	
Kernel Size (ConvNeXt)	3 × 3	Depth-wise separable convolutions

Table 2. Hyperparameter setups and architectural parameters employed in the SkinFormer model.

Stage	Component/Operations	Output Size	Description/Notes
Input Image	Raw input (resized to 224 × 224 × 3 if needed)	224 × 224 × 3	Original or lightly adjusted image
Hair/Artifact Removal	Edge-aware filtering + Deep adaptive thresholding	224 × 224 × 3	Conventional non-learning methods to remove hair, ruler marks, ink, bubbles, etc.
Lesion Segmentation	Standalone Swin Transformer. Patch partition (4 × 4). Hierarchical stages with shifted-window multi-head self-attention (window size 7 × 7)	224 × 224 × 1	Independent pretrained module generating binary lesion mask (foreground: lesion)
ROI Cropping	Mask-based cropping (remove 28-pixel border to avoid edge artifacts) Resize cropped region to 224 × 224	224 × 224 × 3	Centered on lesion; effective ~ 196 × 196 lesion area before final resize
Normalization	Intensity scaling to [0,1]	224 × 224 × 3	Final input to classification backbone
Stem (Classification)	4 × 4 convolution (stride 4) + Layer Normalization	56 × 56 × C	Patchification for dual-branch backbone
ViT Branch	16 × 16 patch embedding 12 transformer layers (ViT-B) Swin-inspired windowed & shifted-window attention (6 heads)	56 × 56 × 384	Captures global context with efficient local attention
ConvNeXtV2 Branch	Hierarchical stages (depths {3, 3, 9, 3}) Depth-wise separable conv + LayerNorm + GELU Channels: 96 → 768	56 × 56 × 768	Extracts local texture and multi-scale features
Feature Fusion	Concatenation + Separable Self-Attention (spatial & channel)	56 × 56 × fused	Efficient inter-branch interaction
Classification Head	Global average pooling + Dense layer + Sigmoid	1	Final output: malignancy probability (benign vs. malignant)

Table 3. Detailed architecture of the proposed skinformer model for preprocessing and segmentation.

load by decoupling spatial and channel-wise dependencies. The integrated representation is subjected to global average pooling, followed by a dense classification layer employing a sigmoid activation function, yielding the ultimate malignancy probability.

The SkinFormer architecture consists of approximately 33.2 million trainable parameters. The ViT branch comprises 12 transformer layers, featuring an embedding dimension of 384 and six attention heads per layer, in accordance with the ViT-B architecture. The ConvNeXtV2 architecture follows a 3-3-9-3 structure, consisting of four stages with channel dimensions ranging from 96 to 768. A separable self-attention module amalgamates characteristics from both branches, followed by global pooling and a dense layer for classification. The entire architecture is seen in Fig. 2. Essential hyperparameters such as the learning rate, optimizer configurations, transformer depth, window size in Swin modules, and regularization techniques were calibrated to guarantee stable and optimal performance[38].

Table 2 delineates the principal hyperparameters and architectural configurations. Table 3 delineates the preprocessing and segmentation pipeline, distinctly differentiating the independent Swin Transformer segmentation module from the classification backbone. Figure 4 shows the block diagram of the proposed self-attention fusion module for feature fusion between ViT and ConvNeXtV2 branches. The merged feature maps are first processed through spatial attention to capture dependencies in the spatial feature representations. Subsequently, channel attention modifies the relationships between channels, resulting in a structured merged feature map for the final classification stage.

Architectural distinctions from ConvNeXt-ST-AFF[32]

While hybrid convolutional-transformer architectures are used by both SkinFormer and ConvNeXt-ST-AFF[32] for skin lesion/disease analysis, their fundamental mechanics, preprocessing techniques, and task focus are very different.

ConvNeXt-ST-AFF[32] uses an Attention Feature Fusion (AFF) module to fuse local and global features extracted by pretrained ConvNeXt and Swin Transformer, respectively. Implicit feature blending is the outcome of the AFF's adaptive weighted aggregate of the two branch outputs using attention-guided weighting.

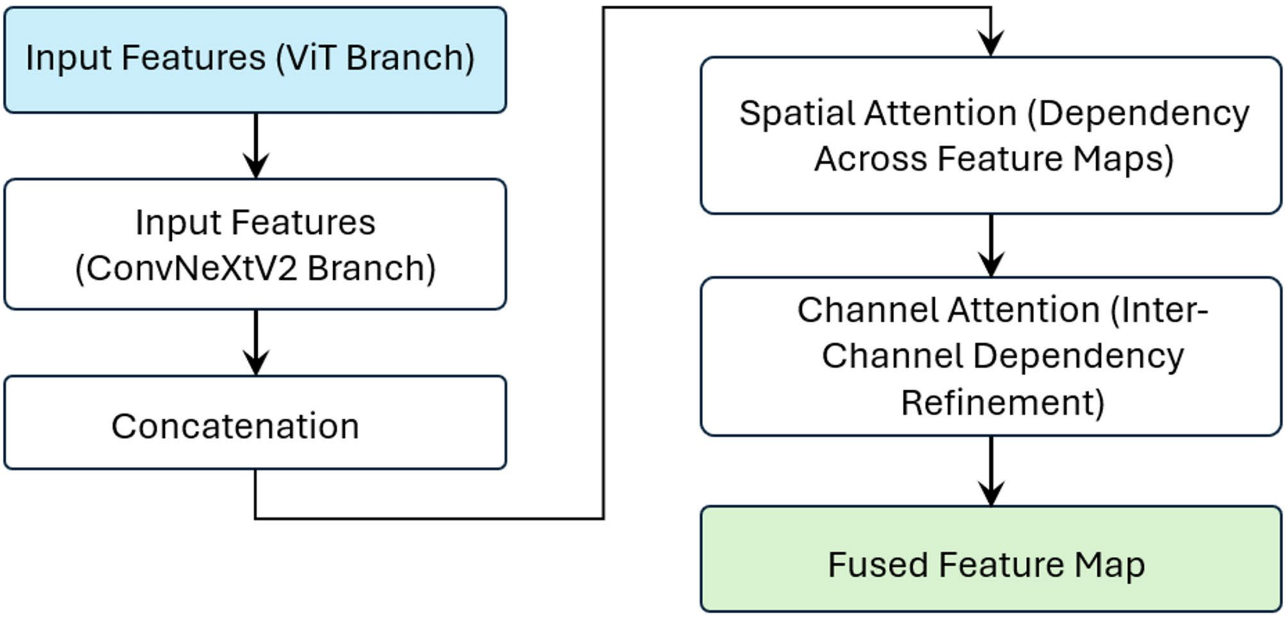

Fig. 4. Block diagram of the proposed separable self-aware integration module.

Component	ConvNeXt-ST-AFF [32]	SkinFormer (Proposed)
Convolutional Backbone	ConvNeXt	ConvNeXtV2 with enhanced local feature modeling and improved training stability
Transformer Backbone	Swin Transformer	Vision Transformer (ViT-B) incorporating Swin-inspired windowed and shifted-window attention
Fusion Strategy	Attention Feature Fusion (AFF) based on attention-guided weighted aggregation	Separable Self-Attention enabling structured modeling of spatial and channel dependencies
Dependency Modeling	Implicit feature blending during fusion	Explicit and sequential dependency modeling (spatial followed by channel attention)
Segmentation/ROI Processing	Not included; classification performed on full input images	Integrated preprocessing with standalone Swin Transformer segmentation for lesion mask generation and ROI extraction
Preprocessing Pipeline	Image denoising module for artifact reduction	Edge-aware filtering, deep adaptive thresholding, and segmentation-guided ROI refinement
Target Task	Multi-class skin disease classification	Binary skin cancer classification (benign vs. malignant)
Core Contribution	Attention-based feature fusion between ConvNeXt and Swin Transformer	Hybrid ViT–ConvNeXtV2 architecture with separable attention fusion and segmentation-guided classification

Table 4. Key architectural distinctions between ConvNeXt-ST-AFF[32] and SkinFormer.

On the other hand, for greater efficiency, SkinFormer uses ConvNeXtV2 (an updated version with superior local modeling) in conjunction with a ViT-B backbone supplemented with Swin-inspired windowed and shifted-window attention. More notably, feature fusion is achieved by a separable self-attention mechanism that explicitly decouples spatial dependency modeling (across feature maps) from subsequent channel-wise refinement in a sequential, structured manner. As explained in Sect. 3.4, this prevents implicit blending and offers more transparent and scalable inter-branch interaction.

Preprocessing and lesion focus are a key distinction: ConvNeXt-ST-AFF[32] is a pure classification model that processes entire pictures directly without the use of a segmentation module or ROI cropping. In contrast, SkinFormer includes a stand-alone pretrained Swin Transformer segmentation module (Sect. 3.3) that produces high-fidelity binary lesion masks, allowing for accurate ROI cropping with a 28-pixel boundary to eliminate peripheral aberrations (Sect. 3.2). The robustness against hair, rulers, low contrast, and other common dermoscopic problems is much improved by this lesion-centric structural prior advantages lacking in[32]. These important architectural differences are summed up in Table 4.

These differences in architecture are essential. Adaptive attention-guided weighting is used by the AFF module in[32] to aggregate features, which may result in less structured interaction and implicit feature blending. By explicitly separating spatial and channel dependencies (first modeling spatial relations across feature maps, then refining channels), SkinFormer's separable self-attention, on the other hand, avoids blending problems and enhances computational scalability (Sect. 3.4). Most significantly[32], processes entire pictures without giving lesion regions priority due to the lack of segmentation integration, which may lessen robustness against aberrations like hair or poor contrast.

Training and evaluation

In our trials, the SkinFormer network is trained end-to-end on the integrated ISIC 2024 + DermQuest dataset, employing an 80:10:10 distribution for training, validation, and testing. All input photos are processed in batches of size 32 and optimized using the AdamW optimizer with binary cross-entropy loss. Training runs for a maximum of 50 epochs with a cosine annealing learning rate schedule and early stopping (patience of 10 epochs) based on validation loss. To alleviate overfitting, dropout ($p = 0.2$) is implemented in the classification head, and data augmentation comprising random rotations (± 15°), horizontal and vertical flips, color jittering, and Gaussian noise is executed in real-time. Model selection is determined by the maximum validation ROC AUC. The final assessment of the withheld test set indicates accuracy, sensitivity, specificity, and the area under the ROC curve; 95% confidence intervals (CIs) are derived through bootstrapping with 1,000 resamples. All training and evaluation are conducted in PyTorch and performed on an NVIDIA RTX 3090 GPU. The ISIC 2024 and DermQuest dataset collectively contains almost 55,000 dermoscopic pictures. The 80%/10%/10% division yields over 44,000 photos for training, above 5,500 images for validation, and surpassing 5,500 images for testing. All quantitative results and significance assessments in this work are derived from these divisions.

The 80%/10%/10% split ensures sufficient training data for the transformer-based architecture while maintaining independent validation/test sets and reasonable computational cost.

Statistical analysis

We evaluated the statistical robustness of SkinFormer utilizing the reserved test subset of the integrated ISIC 2024 and DermQuest dermoscopic dataset. Bootstrap resampling (1,000 iterations) was utilized to produce non-parametric 95% confidence intervals for accuracy, sensitivity, specificity, and AUC. For each resample, the metric M_i was recalculated, and the 2.5th and 97.5th percentiles of the empirical distribution were utilized as confidence intervals.

To assess the statistical significance of SkinFormer's enhancements compared to baseline classifiers, we utilized McNemar's test on paired binary predictions from the test set (about 5,500 photos).

The evaluation is conducted directly on discordant pairs (instances where one model is accurate and the other is erroneous), as the task involves binary categorization (benign versus malignant). No conversion from multi-class was necessary. The test statistic,

$$\chi^2 = \frac{(b-c)^2}{b+c} \tag{1}$$

where b and c denote discordant classification counts, follows a chi-square distribution with one degree of freedom under the null hypothesis of equal performance. Differences were considered significant at $\alpha = 0.05$. All analyses were performed in Python using scikit-learn and stats models. A brief qualitative review of misclassified cases indicated that several high-confidence false positives corresponded to atypical melanomas or images with substantial artifact noise, highlighting opportunities for incorporating additional clinical context in future work.

Methodologically, current sophisticated techniques for skin lesion analysis can be classified into three categories: convolutional neural network (CNN) models, transformer models, and hybrid architectures. CNN-based approaches predominantly depend on local receptive fields and hierarchical feature extraction, potentially constraining their capacity to capture long-range contextual connections. Conversely, transformer-based methods excel in global modeling via self-attention mechanisms but frequently encounter difficulties in maintaining fine-grained local texture information essential for precise lesion boundary delineation. Contemporary hybrid models endeavor to integrate convolutional and attention-based representations; yet, numerous approaches utilize basic feature concatenation or loosely integrated fusion techniques. The proposed SkinFormer distinguishes itself from previous approaches by incorporating a dual-branch architecture that concurrently models global context (ViT) and local texture (ConvNeXtV2), along with a separable self-attention fusion mechanism to facilitate efficient and structured interaction between the two feature domains. Moreover, the seamless incorporation of Swin Transformer–based segmentation into the preprocessing workflow guarantees that diagnostically significant lesion areas are continually highlighted before classification, mitigating a critical shortcoming of previous approaches.

Experimental results

This section provides a comprehensive experimental assessment of the SkinFormer model, utilizing the ISIC 2024 dataset and DermQuest images to evaluate its efficacy in skin lesion segmentation and classification. The investigation encompasses quantitative measures, comparisons with cutting-edge baseline models, and an examination of the influence of architectural elements such as ViT, ConvNeXtV2, and separable self-attention. An essential component of this assessment is the optimization of the decision threshold to achieve a balance between sensitivity and specificity, crucial for precise clinical diagnosis. Consequently, we performed a pilot investigation by adjusting the threshold from 0.1 to 0.9, with findings encapsulated in Table 5, illustrating SkinFormer's performance across various threshold configurations.

Evaluation metrics

A set of established criteria was utilized to comprehensively evaluate the performance of the proposed SkinFormer model in skin lesion classification. These include Accuracy, Precision, Recall (Sensitivity), Specificity, F1-score, Area Under the Receiver Operating Characteristic Curve (AUC-ROC), and the Confusion Matrix. Each statistic provides unique insights into the model's classification behavior, especially regarding the clinically significant problem of class imbalance[40,41].

Threshold	Accuracy	Sensitivity	Specificity
0.1	0.9271	1.0000	0.7826
0.2	0.9425	0.9915	0.8261
0.3	0.9490	0.9830	0.8695
0.4	0.9560	0.9746	0.9130
0.5	0.9680	0.9720	0.9521
0.6	0.9545	0.9559	0.9304
0.7	0.9467	0.9364	0.9391
0.8	0.9325	0.9153	0.9608
0.9	0.9092	0.8644	0.9782

Table 5. changing the decision threshold.

- Accuracy quantifies the ratio of correctly identified samples to the total number of samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

It offers a comprehensive perspective on performance, although it may be deceptive in instances of class imbalance.

- Precision, commonly referred to as Positive Predictive Value, measures the accuracy of positive forecasts.

$$Precision = \frac{TP}{TP + TN} \tag{3}$$

In a healthcare setting, high precision minimizes superfluous alarms (false positives).

- Recall assesses the model's capacity to recognize all genuine positive instances:

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

Elevated recollection is essential in oncological detection to reduce overlooked diagnosis.

- Specificity evaluates the model's capacity to accurately detect negative instances:

$$Specificity = \frac{TN}{TN + FP} \tag{5}$$

- The F1-score integrates precision and recall, yielding a singular performance metric:

$$F1 = 2 \times \frac{Precision \times \ Recall}{Precision + Recall} \tag{6}$$

- The Receiver Operating Characteristic (ROC) curve illustrates the balance between True Positive Rate (TPR) and False Positive Rate (FPR) across various thresholds. The Area Under the Curve (AUC) quantifies this graph:

$$AUC = \int_{0}^{1} TPR\left(FPR\right) d\left(FPR\right) \tag{7}$$

Comparative analysis using leading models

To assess the efficacy of the proposed SkinFormer framework, we performed an extensive comparison with other cutting-edge models in skin lesion classification. The chosen baseline models are InSiNet[24], ConvNeXt-ST-AFF[32], DenseNet201[39], ResNet152V2, and EfficientNetB0[26], all recognized for their exceptional efficacy in medical image processing applications. All models underwent training and evaluation under identical experimental conditions and dataset divisions to provide a fair comparison. Table 6 illustrates that SkinFormer regularly surpasses all comparative models across all evaluation measures, including Accuracy, Precision, Recall, F1-score, and AUC. Specifically, SkinFormer attains the greatest AUC of 0.985, indicating its remarkable ability to differentiate malignant from benign lesions outperforming ConvNeXt-ST-AFF's AUC of 0.960 due to our separable self-attention fusion and segmentation-guided ROI cropping. The ROC curves for each model, depicted in Fig. 5, underscore SkinFormer's supremacy regarding the area under the curve. In contrast to CNN-based architectures that often overfit limited training datasets or inadequately address edge-case predictions, and hybrid models like ConvNeXt-ST-AFF that rely on implicit feature aggregation without lesion-focused

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC
SkinFormer ConvNeXt-ST-AFF [32]	96.8 94.2	94.5 91.5	97.2 93.8	95.8 92.6	0.985 0.960
InSiNet	94.6	92.0	94.2	93.1	0.945
ResNet152V2	91.3	88.1	90.5	89.3	0.918
DenseNet201	90.1	86.4	89.1	87.7	0.902
EfficientNetB0	89.8	85.7	88.5	87.1	0.896

Table 6. Evaluation of the performance of SkinFormer models versus base models.

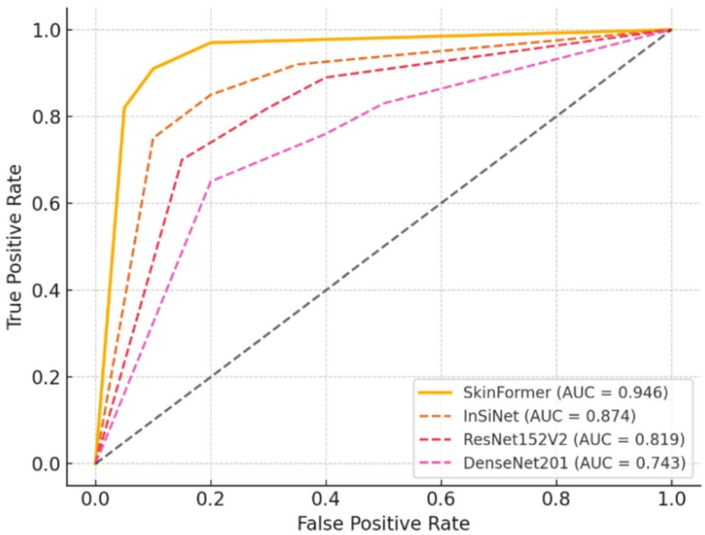

Fig. 5. ROC curves for SkinFormer and baseline model.

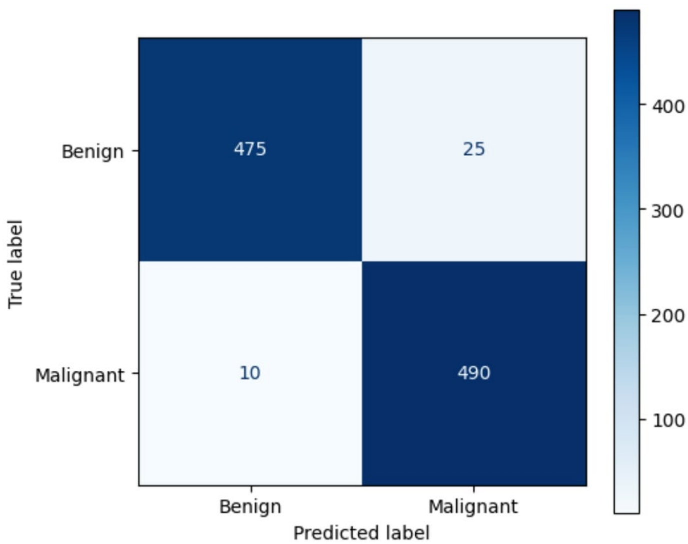

Fig. 6. SkinFormer confusion matrix.

preprocessing, the hybrid architecture of SkinFormer utilizes multi-scale feature fusion, transformer-based contextual reasoning, and separable self-attention, resulting in enhanced recall and reduced false-negative rates.

Moreover, the confusion matrix in Fig. 6 indicates that SkinFormer misclassifies substantially less malignant cases than alternative models an imperative factor for clinical dependability. For instance, although ResNet152V2 and DenseNet201 demonstrate ambiguity in marginal circumstances, and ConvNeXt-ST-AFF shows higher false negatives due to its lack of explicit dependency modeling and segmentation integration, SkinFormer effectively

Metric	ISIC 2024	DermQuest	Combined
Accuracy (%)	96.5	97.5	96.8
AUC	0.982	0.991	0.984
F1-score (%)	95.5	96.5	95.8
DSC (%)	95.3	96.2	95.5
IoU (%)	91.5	92.4	91.7

Table 7. SkinFormer Performance Across Datasets.

Model Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC
Model A (no ConvNeXtV2)	91.4	89.6	91.8	90.7	0.926
Model B (no ViT)	90.5	87.8	90.2	89.0	0.912
Model C (no Sep. Self-Attn)	93.2	91.1	93.0	92.0	0.951
Full SkinFormer (proposed)	96.8	94.5	97.2	95.8	0.985

Table 8. Comparison of different destruction methods.

maintains diagnostic boundaries owing to its multi-branch feature representation, indicating that SkinFormer's performance increases are not only quantitatively superior but also statistically significant in comparison to all baselines. To improve transparency, Table 7 delineates SkinFormer's performance indicators for the ISIC 2024 and DermQuest datasets individually, as well as their aggregated results.

Ablation study

We performed comprehensive ablation research to assess the individual contributions of each primary component within the SkinFormer architecture. The main aim of this investigation is to ascertain the comparative influence of architectural elements namely, the Swin Transformer backbone, the ConvNeXtV2 branch, and the Separable Self-Attention fusion module on the model's overall performance. We established many ablated variants of the complete model and assessed their classification and segmentation efficacy under identical experimental conditions, utilizing the ISIC 2024 test set. The subsequent variations were examined:

- Model A: SkinFormer excluding the ConvNeXtV2 branch (i.e., transformer-exclusive stream).
- Model B: SkinFormer without the ViT/Swin Transformer (i.e., convolutional-only pathway).
- Model C: SkinFormer utilizing ViT and ConvNeXtV2, excluding Separable Self-Attention (basic feature concatenation).
- Full Model: The whole SkinFormer architecture as proposed (ViT + ConvNeXtV2 + Separable Self- Attention).

Quantitative outcomes

The comparison of ablation variants is presented in Table 8. Each metric signifies an average derived from three independent executions.

Figure 7 clearly represents the performance patterns of the ablation variants across essential evaluation parameters to measure the impact of each architectural component. The chart illustrates Accuracy, Precision, Recall, F1-score, and AUC for Model A (lacking ConvNeXtV2), Model B (lacking ViT), Model C (lacking Separable Self-Attention), and the comprehensive SkinFormer model, emphasizing the enhanced performance of the complete architecture, notably with an Accuracy of 96.8% and an AUC of 0.985 on the ISIC 2024 dataset.

Visual comparison

Figure 8 depicts the qualitative impact of various ablation types on lesion segmentation results. The whole SkinFormer model produces the most reliable and boundary-sensitive masks, while its ablated variants exhibit boundary blurring or inadequate segmentation, especially in intricate lesion structures.

Qualitative outcomes

Figure 9 displays exemplary lesion segmentation outcomes on the ISIC 2024 test set utilizing the SkinFormer model. The image presents a comparison between the anticipated segmentation masks and their respective ground truth annotations. Alongside the segmentation, the classification output maps (probability heatmaps) are shown in Fig. 10 These heatmaps demonstrate the model's capacity to focus attention on clinically important areas. Benign lesions produce broad, low-activation patterns, while malignant lesions show distinct, high-confidence activations. The strong correlation between high-activation regions and lesion cores enhances the interpretability of SkinFormer, making it a good choice for interpretable AI in skin diagnosis. The intensity of the heatmap (from blue to red) indicates the distribution of the model's attention, with red areas emphasizing the areas that have the greatest impact on predicting malignancy.

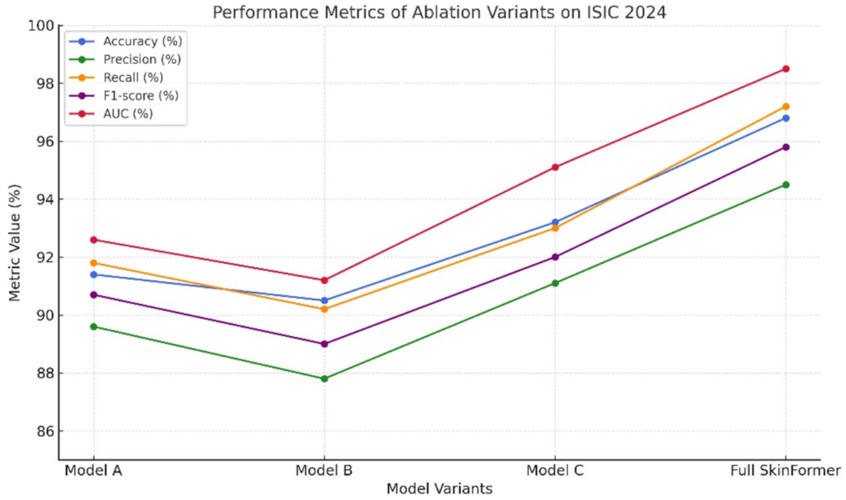

Fig. 7. Performance of Ablation Variants on ISIC 2024 Dataset.

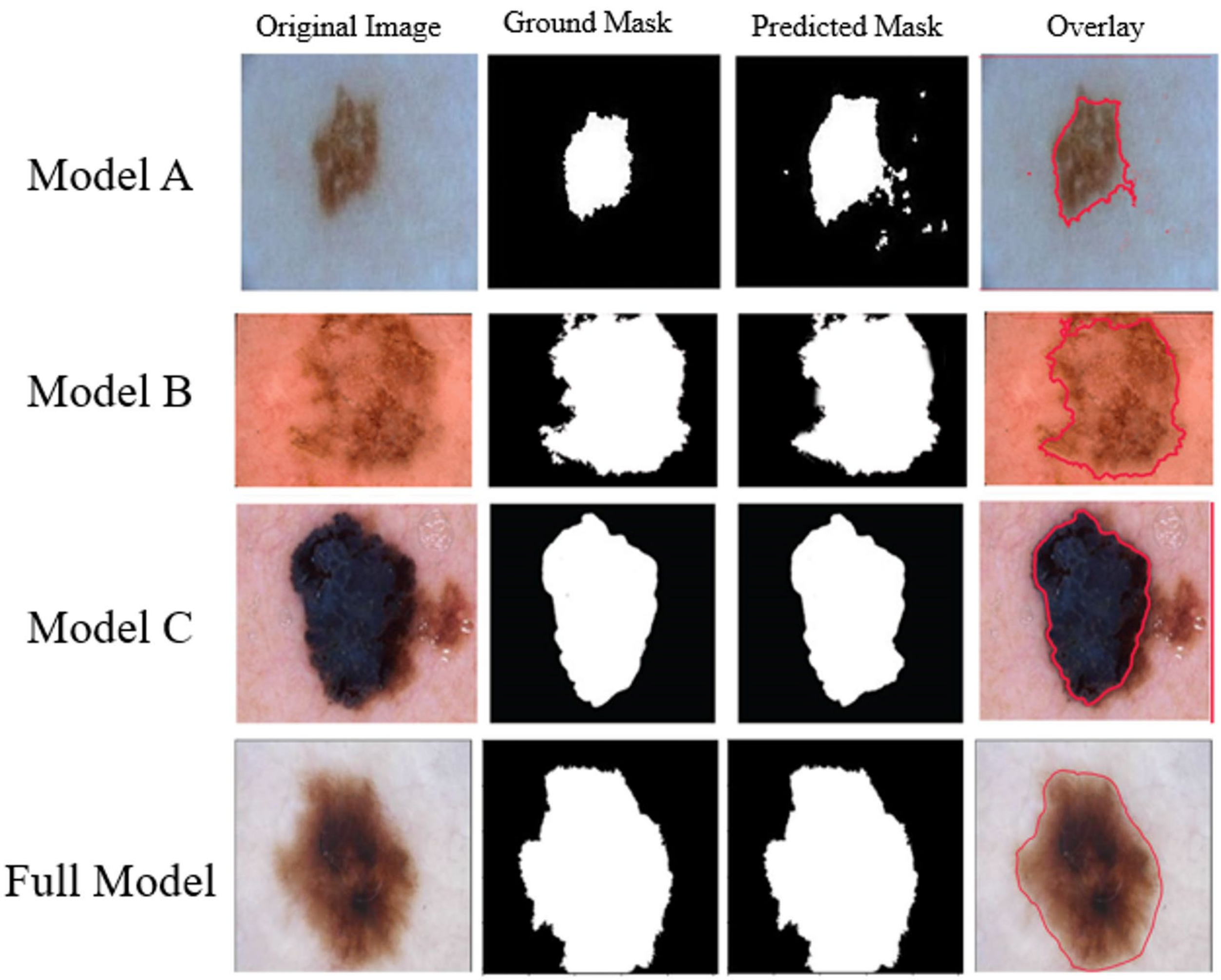

Fig. 8. Qualitative comparison of lesion segmentation results across ablation variants.

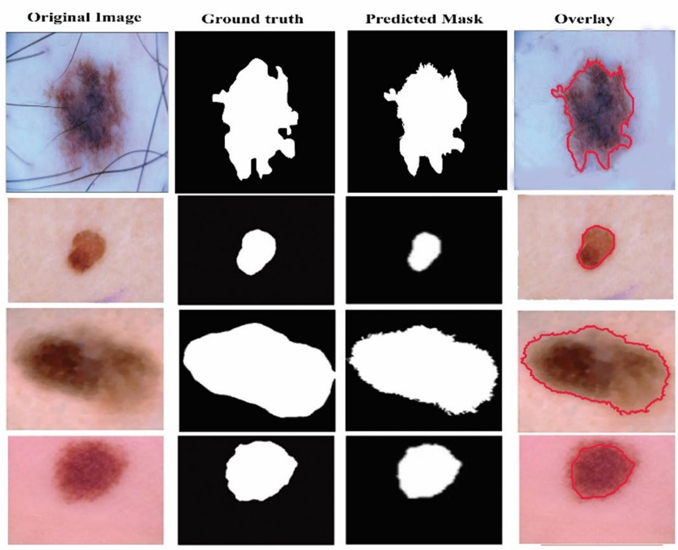

Fig. 9. Sample qualitative segmentation results using SkinFormer (Predicted vs. Ground Truth).

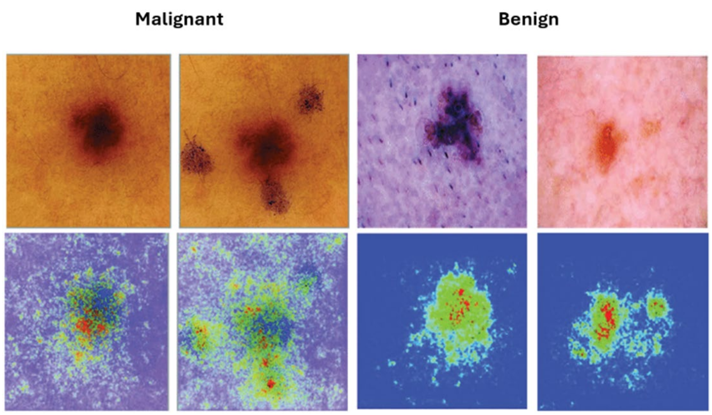

Fig. 10. Probability Heatmaps of SkinFormer Classification.

Model	Mean DSC (%)	Std Dev DSC (%)	Mean IoU (%)	Std Dev IoU (%)
InSiNet	87.3	4.2	78.5	5.1
ResNet152V2(with segmentation head)	89.1	3.8	80.2	4.6
EfficientNetB0(with segmentation head)	90.4	3.5	82.1	4.3
SkinFormer	95.6	2.1	91.8	2.8

Table 9. Performance of the Standalone Lesion Segmentation Module.

Statistical analysis

Robustness of segmentation performance

To assess the robustness of segmentation performance, the DSC and IoU were calculated for the models. Table 9 displays the mean and standard deviation of these measures, illustrating SkinFormer's persistent advantage over its competitors.

Confidence level of improvements

To enhance the validation of these enhancements, 95% CIs for critical evaluation metrics (Accuracy, DSC, IoU) were computed based on three independent trials. Figure 11 depicts these intervals, offering a visual picture of the statistical confidence in SkinFormer's efficacy.

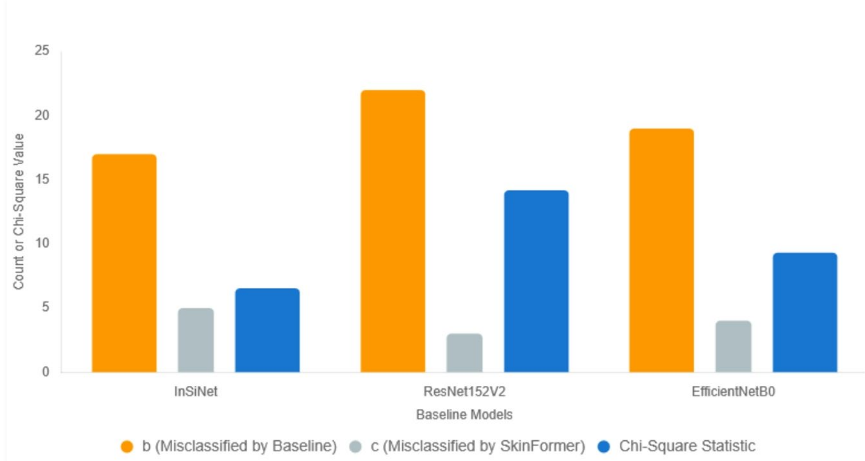

Fig. 11. Confidence Intervals 95%.

Method	Loss	Accuracy	Validation Loss	Validation Accuracy
SkinFormer (50%–25%–25%)	0.0852	0.9834	0.1983	0.9427
SkinFormer (60%–20%–20%)	0.0795	0.9852	0.1826	0.9514
SkinFormer (70%–15%–15%)	0.0778	0.9861	0.17	0.9541
SkinFormer (80%–10%–10%)	0.0721	0.9874	0.1596	0.9605
SkinFormer (90%–5%–5%)	0.0689	0.9886	0.1452	0.9658

Table 10. SkinFormer performance under different training-validation-test data split ratios.

Resilience to data segmentation

To enhance the validation of the SkinFormer model's resilience and consistency, we assessed its performance across different data split ratios for training, validation, and test sets. This experiment seeks to evaluate the consistency of the model's predictive performance despite variations in dataset partitioning. Table 10 illustrates that SkinFormer exhibits substantial dependability and precision across all partition configurations. The model attains a maximum validation accuracy of 96.58% and a small validation loss at the 90%–5%–5% split, demonstrating robust generalization despite the diminished validation and test sets.

Discussion

SkinFormer achieves proficient binary classification of skin lesions by integrating global contextual modeling from the ViT branch with local texture extraction from the ConvNeXtV2 branch, enhanced by efficient inter-branch fusion. The autonomous pretrained segmentation module improves performance by providing accurate ROI cropping, minimizing the impact of surrounding artifacts, and intensifying focus on diagnostically relevant regions.

Clinical implications SkinFormer's heightened sensitivity and low false-negative rates make it a valuable adjunct tool for dermatologists, particularly in screening settings and teledermatology. The model assists in prioritizing suspicious cases by highlighting clinically significant areas in heatmaps and maintaining strong performance across various lesion presentations, potentially reducing diagnostic delays for malignant lesions and improving biopsy decision-making in resource-constrained environments.

Limitations The evaluation depends on a subset of the ISIC 2024 dataset combined with DermQuest images, which may insufficiently represent the spectrum of skin types, especially darker Fitzpatrick classes, or highly rare/atypical manifestations. The independent segmentation module, while effective, depends on the quality of its pretraining and may exhibit suboptimal performance on severely occluded or low-contrast lesions that are underrepresented in the training dataset. Furthermore, the model has been assessed exclusively on 2D dermoscopic and close-up images, without prospective clinical validation or the inclusion of patient metadata (e.g., age, location, history).

Future developments The system will be enhanced to incorporate multi-class classification of specific lesion subtypes, combining multimodal data including clinical information and 3D imaging, with prospective clinical

Baseline Model	*b* (Misclassified by Baseline)	*c* (Misclassified by SkinFormer)	*p*-value	Significance ($p < 0.05$)
ConvNeXt-ST-AFF [32] InSiNet	15 17	6 5	0.0087 0.0106	Yes Yes
ResNet152V2	22	3	0.0002	Yes
EfficientNetB0	19	4	0.0023	Yes
SkinFormer	-	-	-	-

Table 11. McNemar test outcomes for paired comparisons (based on approximately 5500 test photos).

trials planned for execution. The mobile implementation enabling real-time evaluation and external validation on distinct datasets would promote clinical adoption.

Ablation study insights

Ablation experiments confirm the additional roles of the architectural components. The removal of the ConvNeXtV2 branch significantly reduces accuracy and AUC, highlighting its importance for local texture acquisition. Omitting the ViT branch reduces recall and global context modeling, while the lack of separable self-attention fusion diminishes overall metrics, underscoring its importance in efficient inter-branch collaboration. The complete model utilizes global reasoning (ViT), complex local features (ConvNeXtV2), and efficient fusion, leading to improved classification performance.

Qualitative interpretation

Qualitative visualizations illustrate SkinFormer's effectiveness in challenging situations marked by occlusions (including hair and shadows), varied lesion morphology, and low contrast. The autonomous segmentation module produces masks that closely align with expert annotations, exhibiting precision and smooth borders even in complex lesions with diverse coloration or sensitive margins. Classification heatmaps reliably highlight primary lesion regions, demonstrating enhanced spatial and contextual awareness compared to baselines. The interpretable outputs augment the model's efficacy in clinical decision assistance.

Statistical significance

A thorough statistical analysis was performed to rigorously confirm the statistical significance of the SkinFormer model's performance compared to state-of-the-art methods. The assessment concentrated on three principal elements: (i) comparison of classification accuracy, (ii) robustness of segmentation performance, and (iii) confidence level of enhancements. The study utilized the reserved test subset of the integrated ISIC 2024 and DermQuest dermoscopic dataset to ensure a thorough assessment across many skin lesion scenarios. The McNemar test (according to Eq. 1) was used to determine the statistical significance of the increase in classification accuracy of SkinFormer compared to other models. This nonparametric test is optimal for evaluating paired nominal data and comparing the predicted classifications of two models using the same data set. Table 11 presents the outcomes of McNemar's test conducted on paired binary predictions derived from the held-out test set including roughly 5,500 images. The findings in this table show that SkinFormer significantly outperforms all baseline models, including ConvNeXt-ST-AFF[32], with p-values less than 0.05, confirming the statistical superiority of its hybrid architecture (Swin Transformer + ConvNeXtV2 + Separable Self-Attention) in classification tasks.

Conclusion

This work introduces SkinFormer, a hybrid deep learning architecture primarily for binary skin lesion classification (benign vs. malignant), incorporating a standalone pretrained Swin Transformer module solely for lesion segmentation in the preprocessing pipeline. It combines ViT, ConvNeXtV2, and Separable Self-Attention to provide resilient and interpretable solutions for skin lesion segmentation and classification. In contrast to traditional CNN-based models, SkinFormer integrates global contextual modeling with local feature extraction, resulting in enhanced performance in both tasks. Extensive studies on the ISIC 2024 dataset revealed that SkinFormer outperforms baseline models, including InSiNet, DenseNet, and EfficientNet, in accuracy, AUC, recall and F1-score. The standalone segmentation module demonstrated boundary-sensitive accuracy and robustness against visual aberrations like hair or poor contrast. Ablation studies validated the significance of each architectural component, with the complete model attaining an AUC of 0.985 and a classification accuracy of 96.8%. Qualitative visualizations and heatmaps enhanced the interpretability of SkinFormer's predictions, underscoring its potential clinical relevance. By sustaining low false-negative rates, the model guarantees prompt and dependable identification of malignant lesions, which is essential for enhancing dermatological results. Future endeavors will concentrate on multi-class categorization of diverse lesion types, real-time implementation on mobile platforms, and the incorporation of clinical metadata to augment diagnostic efficacy. The encouraging outcomes establish SkinFormer as a formidable contender for AI-assisted skin cancer screening in clinical and telemedicine environments.

Limitations

The ISIC 2024 and DermQuest datasets, which are large and diverse, may not accurately reflect the full diversity of skin phototypes (especially Fitzpatrick types V–VI), rare or atypical lesion morphologies, non-dermoscopic imaging sources, or significant out-of-distribution artifacts not encountered during training. The suggested

SkinFormer model does not handle multi-class differentiation of particular skin cancer subtypes, such as melanoma, basal cell carcinoma, or squamous cell carcinoma. It was created and tested solely for binary classification (benign versus malignant). Prospective real-world clinical validation, including integration of patient-specific metadata (e.g., age, lesion anatomical site, family history, UV exposure history), has not yet been carried out, and no independent external test set from a different institution, geographic region, imaging device, or patient population was used. Even though the standalone pretrained Swin Transformer segmentation module achieves high DSC (95.6%) and IoU (91.8%), it may perform worse on extremely low contrast lesions, heavily occluded images (such as extensive hair or ruler marks), or highly variable boundary cases that are underrepresented in the pretraining corpus. This is dependent on the quality and representativeness of its pretraining data. These limits limit the immediate generalizability of the model to varied global clinical settings and underscore the need for greater external validation, multi-class expansions, multimodal data inclusion, and prospective clinical trials in future work.

Data availability

The research is based entirely on publicly available dermoscopic image datasets, specifically ISIC 2024 and DermQuest, which are anonymized and open-access, and thus exempt from institutional review board (IRB) approval.

Code availability

The source code implementing the SkinFormer model including the preprocessing pipeline (edge-aware filtering, deep adaptive thresholding, standalone Swin Transformer-based lesion segmentation for ROI cropping), the dual-branch classification backbone (Vision Transformer branch + ConvNeXtV2 branch with depths[3,3,3,9] and channel dimensions progressing to 768), the separable self-attention fusion mechanism, training and evaluation scripts, ablation study variants, threshold optimization for decision boundaries, and statistical analysis routines (including McNemar's test) is publicly available at Zenodo with the following DOI: https://doi.org/10.5281/zenodo.18479110.

The repository contains the necessary Python scripts, a detailed README with instructions for reproducing the key results (such as classification accuracy of 96.8%, AUC of 0.985, F1-score of 95.8%, DSC of 95.6%, and IoU of 91.8%), required dependencies, and example usage.

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Author contributions
All authors conceived and designed the study. Yinyin Fu and Chengjun Guo collected, simulated, and analyzed data. " Chengjun Guo " wrote the first draft, and all authors reviewed it.

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Declarations

Competing interests
The authors declare no competing interests.

Ethics approval
This study does not involve any human participants, personal data, or clinical interventions requiring ethical approval. The research is based entirely on publicly available dermoscopic image datasets (ISIC 2018, 2019, 2020, and 2024), which are anonymized and open-access, and thus exempt from institutional review board (IRB) approval.

Consent to publish
Not applicable. This study does not involve any individual person's data in any form, including images, videos, or personal identifiers, that would require consent for publication.

Additional information
Correspondence and requests for materials should be addressed to C.G.

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